Philosophical Foundations of AI Ethics

Roadmap

Introduction

Moral Philosophy and Ethical Frameworks

Utilitarian, Deontological, and Virtue Ethics in AI Context

Case Studies

Introduction

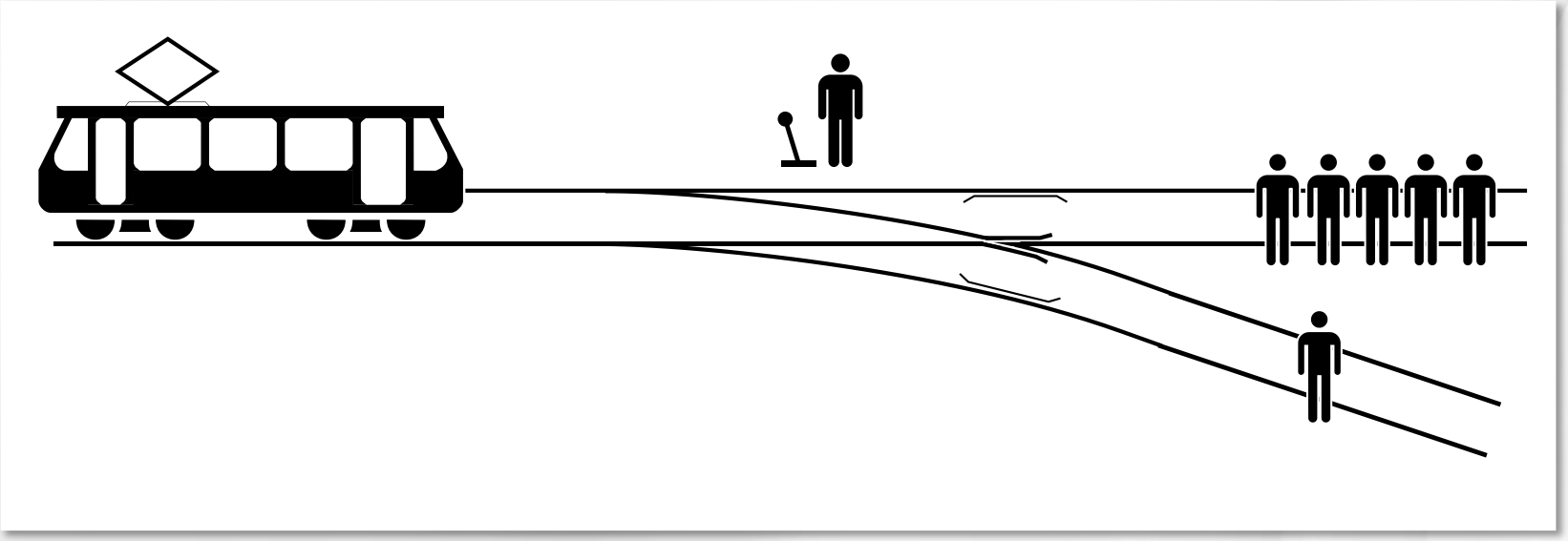
Early Discussions of Machine Ethics

Early discussions traced back to

- Science fiction works (e.g., Isaac Asimov's Three Laws of Robotics)
- Early days of computing (e.g., Norbert Wiener's cybernetics)

Trolley Problem

A series of thought experiments involving ethical dilemmas



https://en.wikipedia.org/wiki/Trolley_problem

Ethical Dilemmas

- Ethical dilemmas are situations in which a person faces conflicting moral principles or values when deciding the best course of action between two or more choices
- The ethical framework one adopts significantly influences how these dilemmas are approached and resolved, shaping the decision-making process and the values prioritized in the outcome
- Understanding different ethical frameworks can help individuals navigate complex ethical situations more effectively

Moral Philosophy and Ethical Frameworks

Foundations of Moral Philosophy

- Moral philosophy: examines questions of right and wrong, good and bad, and the nature of moral reasoning and decision-making
- Moral responsibility: refers to the obligation to act ethically and accept consequences of one's actions
- Moral decision-making: evaluates choices and actions based on ethical principles

Ethical Frameworks for AI

- Consequentialist approaches
- Deontological frameworks
- Virtue and care-based ethics
- Social and justice-oriented frameworks
- Principlism and hybrid approaches

Applying Traditional Ethics to AI: Strengths

- Established frameworks provide starting points for AI ethical guidelines
- Utilitarian ethics aligns with AI's data-driven nature for quantifiable outcomes
- Deontological ethics offers clear rules for AI behavior in various situations
- Virtue ethics presents a way to imbue AI systems with desirable traits (e.g., honesty, fairness)

Applying Traditional Ethics to AI: Limitations

- Human-centric theories may not fully apply to non-human AI systems (e.g., consciousness, intentionality)
- AI's speed and scale of operation require adaptations to traditional ethical theories
- Potential superintelligent AI surpassing human cognition may need new ethical frameworks
- Translating human virtues into computational terms poses significant challenges

Addressing Ethical Complexities in AI

- Interdisciplinary collaboration between philosophers, ethicists, and AI researchers needed
- Developing hybrid ethical frameworks combining multiple approaches for AI systems
- Incorporating ethical considerations throughout the AI development lifecycle
- Creating flexible ethical guidelines adaptable to evolving AI capabilities and applications

Utilitarian, Deontological, and Virtue Ethics in AI Context

Core Principles and Frameworks of Ethics

Utilitarian Ethics	Deontological Ethics	Virtue Ethics
• Focuses on outcomes, consequences, goals, ends (emphasize context and flexibility)	• Focuses on standards, compliance, duties, rules, means (emphasize consistency)	• Focuses on values, character, intentions, motives, attitudes

Comparison of Ethical Frameworks in AI

- Key questions asked
  - Utilitarian ethics: “Does it produce net good?”
  - Deontological ethics: “Does it follow the rule?”
  - Virtue ethics: “Does it contribute to virtue?”
- Often used together to balance innovation with risk, fairness with efficiency, and short-term gains with long-term societal consequences

Utilitarian Ethics

- Emphasizes maximizing overall well-being or happiness for greatest number of individuals
- Focuses on consequences of actions rather than actions themselves
- Principle of utility: the most ethical choice produces greatest good for greatest number of people

https://mediaethicsutar.wordpress.com/4-theories/utilitarianism/

Key Components of Utilitarian Ethics

- Consequentialism: evaluates morality based on outcomes
- Hedonistic calculus: attempts to quantify pleasure and pain (Bentham)
- Rule utilitarianism: advocates following rules that generally maximize utility

Strengths and Limitations of Utilitarian Ethics

Strengths

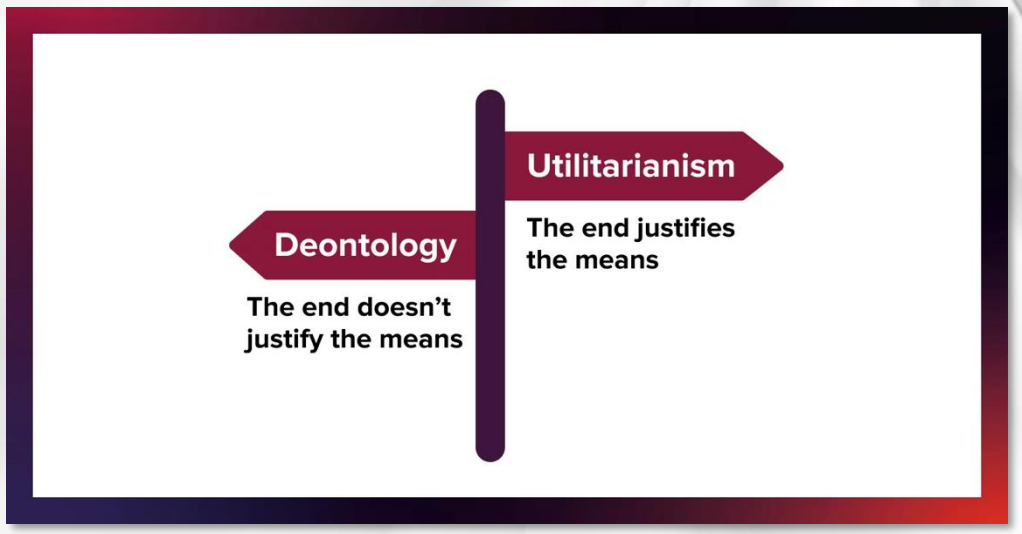
- Well-suited for cost-benefit analysis of AI systems (e.g., healthcare AI improving patient outcomes)
- Adaptable to changing technological landscape and societal needs

Limitations

- Difficulty in quantifying diverse impacts of AI (e.g., social media effects on mental health)
- Risk of prioritizing majority benefits over minority protections

Deontological Ethics

- Judges morality of actions based on adherence to rules or duties
- Emphasizes inherent rightness or wrongness of actions regardless of consequences
- Kant's Categorical Imperative central concept stating to act only according to rules that could become universal laws



https://uditangshu.com/utilitarianism-vs-deontology/

Key Principles of Deontological Ethics

- Moral absolutism: holds certain actions always right or wrong (e.g., lying, stealing)
- Duty-based ethics: focuses on fulfilling moral obligations
- Rights-based ethics: emphasizes respecting individual rights

Strengths and Limitations of Deontological Ethics

Strengths

- Provides clear ethical guidelines for AI development (e.g., IEEE Ethically Aligned Design)
- Protects fundamental rights in face of powerful AI capabilities

Limitations

- May impede beneficial AI developments due to rigid rules
- Struggles with ethical dilemmas in AI decision-making (e.g., autonomous vehicle trolley problems)

Virtue Ethics

- Focuses on moral character of individuals rather than actions or consequences
- Emphasizes cultivation of virtuous traits (e.g., courage, wisdom, justice)
- Concept of eudaimonia (i.e., human flourishing) as central goal of ethical behavior

Deontological vs. Virtue Ethics

Ethics of duty (deontology)

- What should I do?
- What is my obligation?
- What are the universal moral guidelines?

Ethics of virtue (virtue ethics)

- What kind of person should I be?
- What traits of character should I develop?
- How can I be truly happy? (eudaimonia)

Key Components of Virtue Ethics

- Moral exemplars: serve as role models for virtuous behavior
- Practical wisdom (phronesis): guides application of virtues in specific situations
- Virtue cultivation: through habit and practice

Strengths and Limitations of Virtue Ethics

Strengths

- Emphasizes character development of AI practitioners
- Encourages holistic approach to ethical AI design

Limitations

- Challenges in defining and implementing virtues in AI systems
- Lack of clear metrics for evaluating ethical performance of AI

Case Studies

Princeton AI Ethics Case Study 1

Case study on an automated healthcare app: https://aiethics.princeton.edu/wp-content/uploads/sites/587/2018/10/Princeton-AI-Ethics-Case-Study-1.pdf

Group Discussion: Case Anaysis Using Three Different Ethical Frameworks

1. Analyze Princeton case study 1 using each of the three ethical frameworks (i.e., utilitarian ethics, deontological ethics, and virtue ethics) separately, listing specific examples and reasons to argue for or against the case
2. Explore feasible ways to integrate the promising aspects of different ethical frameworks to mitigate the ethical concerns

Remember to take notes during your discussions to facilitate subsequent reporting